

5.2.11. Scatter Plot for Age vs. Fare by Survived

03:35

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina
Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Note: Refer to the visible test case for better reference.

Code:

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Load the Titanic dataset
```

```
data = pd.read_csv('Titanic-Dataset.csv')
```

```
# Data Cleaning
```

```
data['Age'].fillna(data['Age'].median(), inplace=True)
```

```
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
```

```
data.drop('Cabin', axis=1, inplace=True)
```

```
# Convert categorical features to numeric
```

```
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
```

```
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
# Write your code here for Scatter Plot for Age vs. Fare by Survived
```

```
colors=data['Survived'].map({0:'red',1:'blue'})
```

```
plt.scatter(data['Age'],data['Fare'],c=colors)
```

```
plt.title("Age vs. Fare by Survival")
```

```
plt.xlabel("Age")
```

```
plt.ylabel("Fare")
```

```
plt.show()
```

Average time

1.965 s

1965.00 ms



Maximum time

1.965 s

1965.00 ms



✓ 1 out of 1 shown test case(s) passed

✓ Test case 1 1965 ms

Debug



Expected output

Actual output

